

Computer Science Department

REAL TIME FRUIT DETECTION USING DEEP LEARNING



Project Guide:

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Introduction

In today's technology-driven world, real-time fruit detection plays an important role in agriculture and retail industries. It helps in automatically identifying and classifying different fruits from images or live video streams.

By using advanced technologies like deep learning and computer vision, the system detects fruits based on features such as shape, color, and size. Models like CNN and YOLOv8 provide fast and accurate results, reducing manual effort improving efficiency.





Computer Science Project

Objectives

1. To Detect Fruits in Real-Time –

The system detects fruits instantly using a camera or live video.

2. To Classify Different Types of Fruits

The system identifies the type of fruit like apple, banana, mango, etc.

3. To Localize Fruits Using Bounding Boxes

The system draws a box around the fruit to show its position in the image.

4. To Improve Accuracy Using Deep Learning Algorithms

Deep learning algorithms help the system detect fruits more correctly with fewer errors.

5. To Reduce Manual Work

The system automatically detects and sorts fruits, reducing human effort.



Problem Statement



Existing System

- Difficulty in detecting overlapping or partially visible fruits
- high dependency on controlled environments for accurate detection
- slow processing speed for large image datasets
- requires more manual setup and supervision
- cannot easily adopt to new fruit varieties



Proposed System

- capable of detecting fruits in complex and dynamic environments
- Generates clear visual feedback using bounding boxes and labels
- handles multiple fruit detection in a single frame efficiently
- reduces manual work through automation
- Ensures simple interaction with minimal user effort

Methodology

Data Collection

The first step is collecting images of different fruits such as apple, banana, orange pineapple etc.

Data preprocessing

Before training the model, the dataset is cleaned and prepared.

Dataset splitting

The dataset is divided into two parts
Training Dataset and
Testing Dataset .

Feature Extraction using CNN

A Convolutional Neural Network (CNN) automatically extracts important features from fruit images.

Model Training

The CNN model is trained using the training dataset.

Real-Time Detection

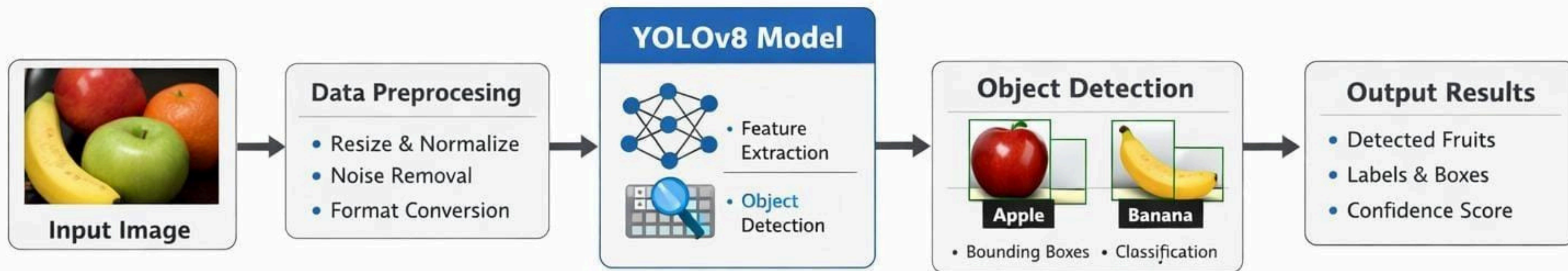
After training, the model is connected with OpenCV to detect fruits in real time.

Output Display

The system shows:
Detected fruit name
Bounding box around the fruit



System Architecture



System Modules

Image acquisition

Captures real-time images or video using a camera.

Data preprocessing

Cleans and resizes images to improve quality and accuracy.

Model training

Trains a model like YOLOv8 to recognize fruits.

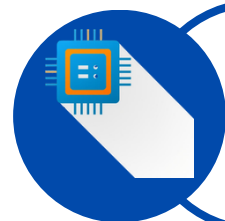
Object detection and classification

Detects fruits and identifies their type (apple, banana, etc.).

Output display

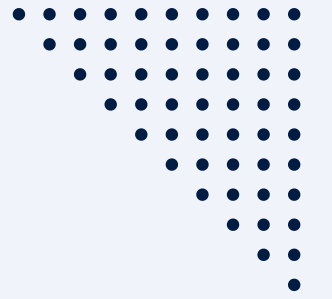
Shows results with fruit name, bounding box, and confidence score.

Tools and Technologies



Hardware Requirements

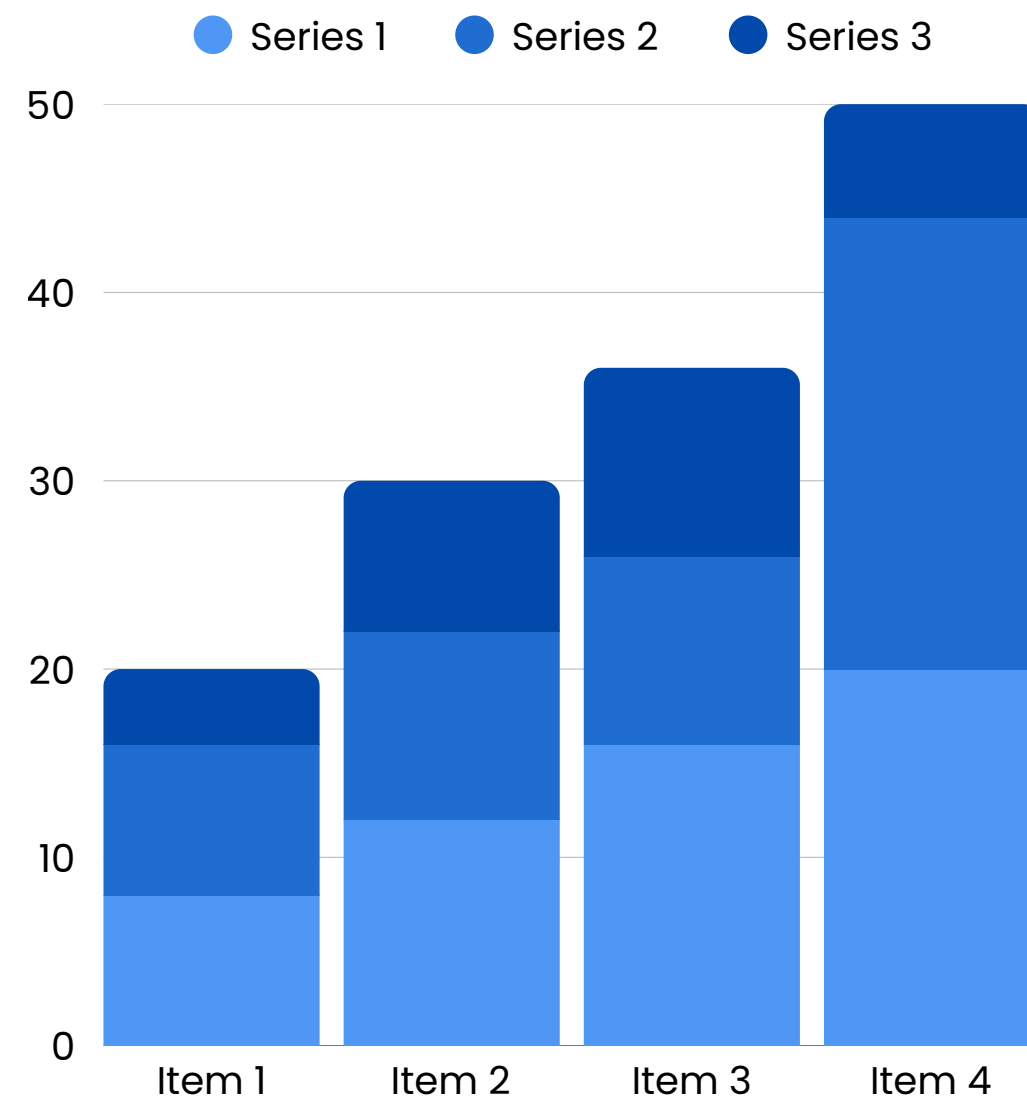
- **System** : intel core i5 processor
- **Hard Disk** : 256GB
- **Monitor** : "15.6" FHD
- **Input Devices** : Keyboard, mouse/touchpad, webcam.
- **RAM** : 8 GB



Software Requirments

- **Operating system** : windows 11
- **Front end** : python
- **Back end** : python(YOLOv8 Deep Learning model)
- **Libraries** : OpenCV, NumPy, Pandas, matplotlib

Expected Outcome



- The system should be able to see and identify different fruits in a video or camera feed.
- When you point the camera at fruits, the system should quickly recognize what type of fruit it is (like apples, bananas, or oranges) and show a box around each fruit on the screen.
- The detection should happen fast and smooth without delays.
- It should work well even with different lighting and angles, and should be able to tell fruits apart even if they are partially hidden or touching each other.
- The whole process should be quick, accurate, and reliable enough to be useful in real applications.

PROJECT ASSOCIATES

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Real-Time Fruit Detection

THANK YOU!

Contact :
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Repository :
github.com/fruit-detect

